KMTalk: Speech-Driven 3D Facial Animation with Key Motion Embedding

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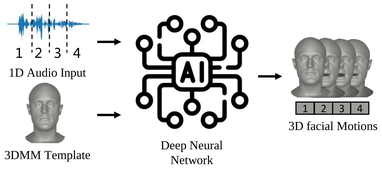
Code: https://github.com/ffxzh/KMTalk

3D Facial Animation

Speech-driven 3D facial animation aims to create realistic 3D Morphable Model sequences that synchronize with input speech.



Challenges

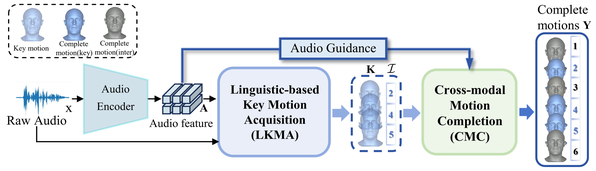
- Main challenge: the ill-posed problem–there may be multiple plausible outputs for input audio,
- Reason: the cross-modal mapping uncertainty from the speech domain to the 3D motion domain.

Motivations

Recent methods map audio signals to 3D facial meshes in a data-driven manner but face several key limitations:

- Sub-optimal temporal coherence and audio-visual consistency cause by directly learning the ambiguous cross-modal mapping,
- Failure to capture detailed facial dynamics and accurate lip movements by overlooking key motion cues,
- Poor lip-speech synchronization from neglecting linguistic priors.

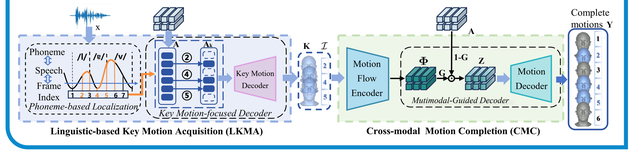
To address these, we introduce a progressive learning mechanism that generates realistic 3D facial animations from audio inputs by incorporating key motion embeddings.



Method Overview

The KMTalk system consists of two modules: LKMA and CMC.

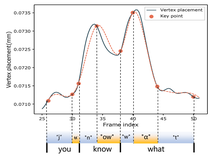
- In the LKMA module, linguistic priors are introduced to locate and generate higher-quality key motions,
- In the CMC module, these key motions are fleshed out into a complete sequence of facial motions.



LKMA

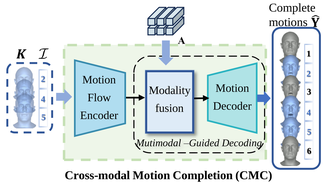
In LKMA module, we leverage linguistic priors to capture pronounced articulatory actions, which are identifiable at phoneme boundaries.

- At phoneme boundaries, a noticeable offset is observed in the articulator movement,
- The *phoneme boundary effects* underscore the ease with which the boundaries of phonemes can be perceived.

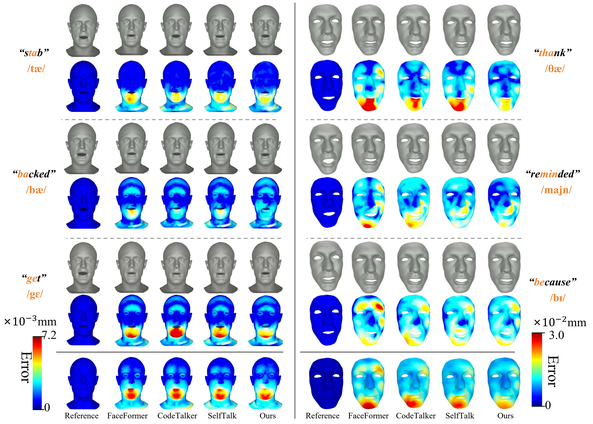


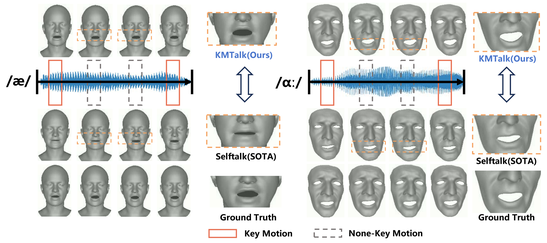
CMC

- While key motions capture essential facial dynamics, they may not encompass all details of non-key motions,
- We jointly combines the audio features, key motions, and key motion indices to generate a complete sequence.



Qualitative Comparisons





- We visualize the output facial meshes from various methods alongside the ground truth for comparison,
- Additionally, the error maps on the left display the vertex L2 loss between the generated and ground truth meshes.

Quantitative Comparisons

Methods	BIWI-Test-A		VOCA-Test	
	LVE↓ ($\times 10^{-4}$mm)	FDD↓ ($\times 10^{-5}$mm)	LVE↓ ($\times 10^{-5}$mm)	FDD↓ ($\times 10^{-7}$mm)
VOCA	6.5563	8.1816	4.9245	4.8447
MeshTalk	5.9181	5.1025	4.5441	5.2062
FaceFormer	5.3077	4.6408	4.1090	4.6675
CodeTalker	4.7914	4.1170	3.9445	4.5422
SelfTalk	4.2485	3.5761	3.2238	4.0912
KMTalk (Ours)	**3.9654**	**2.5446**	**2.2639**	**4.0594**

- The left figure illustrates the competitive results achieved by KMTalk on two widely used datasets, evaluating on the LVE(lip vertex error) and FDD(upper-face dynamics deviation) respectively.
- This indicates that our approach is more effective in achieving audio-visual alignment, thereby leading to improved lip synchronization.

Integration with Existing Methods

Methods		LVE↓ ($\times 10^{-4}$mm)	FDD↓ ($\times 10^{-5}$mm)
FaceFormer	Original	5.3077	4.6408
	After	**5.2793**	**4.2654**
CodeTalker	Original	4.7914	4.1170
	After	**4.5096**	**3.9043**
SelfTalk	Original	4.2485	3.5761
	After	**4.1122**	**2.8668**

- The results of integrating our pro- posed KMTalk with existing methods on BIWI-Test-A.
- Coupling our idea with existing methods consistently improves performance, further verifying the efficacy of our proposed progressive learning mechanism based on key motion acquisition.